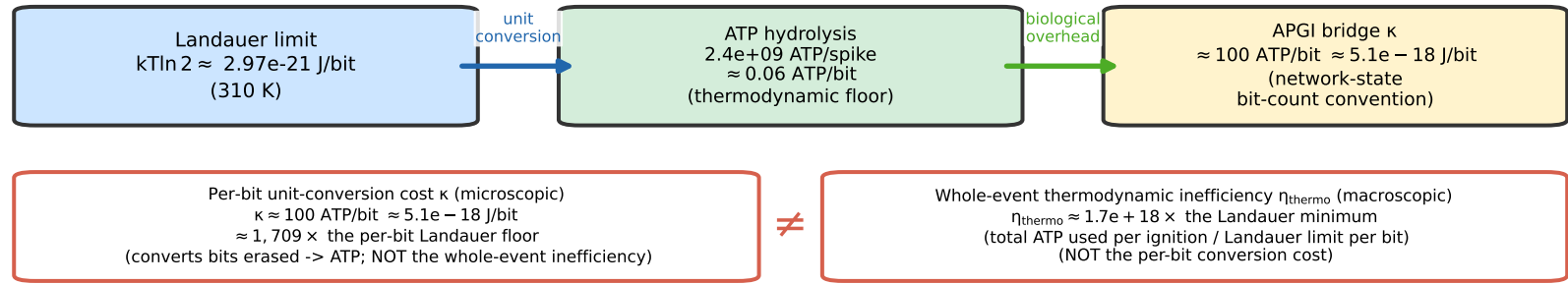


A

$\kappa \approx 100$ ATP/bit is an order-of-magnitude calibration target, not a fitted constant
(conditional on the network-state bit-count convention; falsification window $\kappa \in [10, 1,000]$ ATP/bit)



B

Per-ignition energy budget

- 1. ATP per spike
 $2.4e+09$ ATP
(Attwell & Laughlin 2001; scaled per Lennie 2003)
- 2. Spikes per ignition
 $1e+09$ spikes
($\sim 10^8$ active neurons
 $\times \sim 10$ spikes / ~ 500 ms)
- 3. Total ATP per ignition
 $2.4e+18$ ATP
(order-of-magnitude)

C

Bit-count convention determines κ

- Percept-level coding ($\sim 1-2$ bits/spike)
 $\sim 10^9-10^{10}$ bits
 $\kappa \approx 10^8-10^9$ ATP/bit
- Network-state activation patterns ($\sim 10^{12}$ synapses)
 $\sim 2 \times 10^{16}$ bits
 $\kappa \approx 10^2$ ATP/bit

Six orders of magnitude apart — the $[10, 1,000]$ ATP/bit falsification window is conditional on the network-state convention.